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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct (192344017)

I S. Aswini Sunya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Aswini Sunya

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

1. Describe the evolution from standalone computing to cloud-based system summarize key technological transition that enabled computing
- Computing progressed through several stages before cloud models emerged in the standalone era, each computer operated independently, running its own operating system, storage era, which and application with no shared infrastructure. The next shift was networked client-server computing, multiple machines connected over LANs, so processing and data could be centralized on servers while client accessed them remotely.
- Broadband inter and improved networking which made it practical to access remote computing resources with low latency, rather than relying only on local hardware.
 - Service-oriented architecture and web services.
 - Utility-style, pay-per-use billing models, popularized by providers like Amazon web services in the mid-2000s

Interpret the working of hypervisors in virtual.
Differentiate between type 1 & type 2 hypervisors.

A hypervisor is software that creates and manages virtual machines by abstracting a physical machine's hardware (CPU, memory, storage, network) and allocating it across multiple guest operating systems.

It sits between hardware and the virtual machines,

Intercepting and translating instructions so each

VM believes it has dedicated hardware, while the hypervisor enforces isolation between them.

- Type 1 hypervisors install directly on the physical hardware, with no underlying host operating system. It sits between hardware with no and strong isolation, making them the standard choice in data centers and cloud platforms.

Key diff: Type 1 trades convenience for performance and is used where scale and efficiency matter, while Type 2 trades some performance for ease of use and is better suited to testing, development or as a personal

Describe the role of APIs in cloud service integration
Explain how APIs support interoperability between
services

An application programming interface API is a set
of rules that allows different software applications to communicate
with each other. In cloud computing, APIs enable
users and applications to access cloud resources such as
storage, databases, virtual machines, and networking
services.

Cloud providers offer APIs that allow developers to automate
resource creation and other services, transform data and
integrate applications across different cloud platforms.
APIs provide standard communication methods that allow
applications from different vendors and platforms
to work together. For example, an application
can store files in cloud storage, retrieve data
from a cloud database and send notifications
through another cloud service using APIs.

4. Explain the concept of resource pooling in cloud computing

Resource pooling is a key characteristic of cloud computing in which computing resources such as servers, storage, memory and network are shared among multiple users using a multi-tenant model. The cloud provider dynamically allocates and re-allocates resources based on user demand. Users do not know the exact physical location of the resources but they receive the required services whenever needed.

Benefits of resource pooling:-

- Efficient utilization of hardware resources.
- Reduces operational and infrastructure costs.
- Supports multiple users simultaneously.
- Provides better scalability and availability.
- Simplifies maintenance and management.

Interpret the concept of rapid elasticity in cloud computing

Rapid elasticity is a cloud computing feature that allows computing resources to be automatically increased or decreased based on workload demand. Resources such as CPU, memory, storage and bandwidth can be provisioned or released quickly without manual intervention.

This enables organizations to handle sudden increases or decreases in user traffic efficiently.

How it supports dynamic workloads

- Automatically scales resources during peak demand
- Reduces resources during low usage to save cost
- Maintains application performance and availability
- Eliminates the need to purchase extra hardware
- Supports business continuity and flexibility.

Thus, Rapid elasticity helps organization efficiently manages changing workloads while maintaining performance and minimizing costs.